Grand Quiz Spring 2021

Subject Code MTH301 lecture 1 to 22

Solved By Riz Mughal

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I'm providing 100% correct quiz solution. You can visit my YouTube channel and get more information about all other subjects' quizzes and final year project (CS619).

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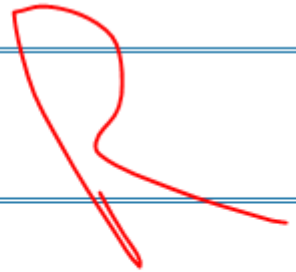
Quiz Start T

Question # 1 of 30 (Start time: 09:21:09 AM, 26 June 2021)

$Face\ of\ the\ parabola\ y = ax^2 + bx + c\ is\ opening\ upward\ when$

Select the correct option

- ○ b > 0
- ○ b < 0
- ○ a < 0
- ● a > 0



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Quiz Start Time: 09:21 AM,

Question # 2 of 30 (Start time: 09:21:29 AM, 26 June 2021)

If $w = x + y$ while $x = r + s$ and $y = r - s$ then which of the following is partial derivative of w w.r.t s?

Select the correct option

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●	0
○	-1
○	2
○	1

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Question # 3 of 30 (Start time: 09:22:23 AM, 26 June 2021)

The surface:z=f(x,y) falls (decreases) most rapidly at any point in the ---------- of Gradient(z).

Select the correct option

- ○ Direction
- ○ Direction perpendicular to
- ○ Arbitrary direction
- ● Opposite to the direction

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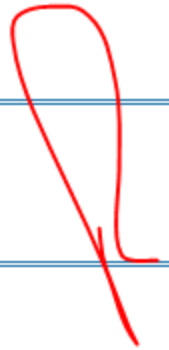
Question # 4 of 30 (Start time: 09:22:48 AM, 26 June 2021)

XY-Plane consist of all points where

Select the correct option

- xy=0
- xy=1
- z=0
- z=1



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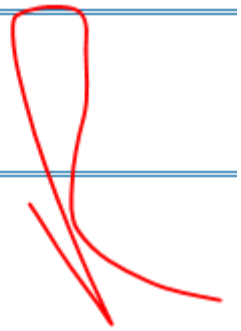
Question # 5 of 30 (Start time: 09:23:03 AM, 26 June 2021)

What is the condition of the extreme value theorem?

Select the correct option

- (●) A function must be continuous on a closed interval *
- ( ) A function must be continuous and differential
- ( ) No conditions are necessary
- ( ) A function must differential



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Quiz Start Time: 09:21

Question # 6 of 30 (Start time: 09:23:19 AM, 26 June 2021)

If $f(x, y, z) = x^2 + 2y - 3z^2$ then f_x, f_y and f_z are

Select the correct option

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- $f_x = 2x, f_y = 2$and $f_z = 6z$
- $f_x = 2x, f_y = 2$and $f_z = -6z$
- $f_x = 2x, f_y = 2y$and $f_z = 6z$
- $f_x = 2x, f_y = 2y$and $f_z = -6z$

Click to Save Answer & Move to

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Question # 7 of 30 (Start time: 09:23:42 AM, 26 June 2021)

Can we evaluate the following integral in given order of integration

$$\int_{0}^{\frac{1}{2}} \int_{2x}^{1} e^{y^2} \, dy dx$$

Select the correct option

○	Yes
●	No

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Quiz Start Time

Question # 8 of 30 (Start time: 09:24:02 AM, 26 June 2021)

The $\int_0^1 2xe^{x^2}\,dx =$

Select the correct option

- $e+1$
- e^{x^2}
- $e-1$ (selected)
- e

Click to Save Answer & M

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Question # 9 of 30 (Start time: 09:24:18 AM, 26 June 2021)

If a vector of magnitude '4' is making angle 30-degree with Y-axis then its component vector along x-axis is --------

Select the correct option

- ○ 2*sqrt(3)
- ○ -2*sqrt(3)
- ● 2
- ○ -2

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Question # 10 of 30 (Start time: 09:24:34 AM, 26 June 2021)

The function

$$f(x,y,z)=\frac{(x^2+yz)}{(xy+z)}$$

is discontinuous when

Select the correct option

- $xy>z$
- $x+y+z=0$
- $xy+z=0$
- $x^2+yz=0$

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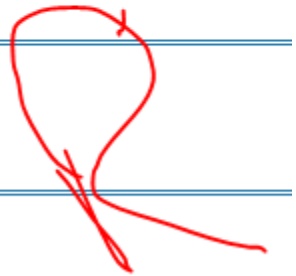
Question # 11 of 30 (Start time: 09:25:05 AM, 26 June 2021)

Value of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at the point $(0, 3, 0)$ is _____

Select the correct option

Re

- ○ $2\sqrt{2}$
- ○ 9
- ● 3
- ○ $\sqrt{3}$



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Question # 12 of 30 (Start time: 09:25:26 AM, 26 June 2021)

$$\int_{1}^{3}\int_{0}^{1} xy\,dx\,dy = \text{- - - - - - - - - -}$$

Select the correct option

- $\frac{3}{4}$
- $\frac{7}{12}$
- $\frac{9}{4}$
- 3

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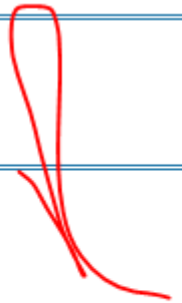
Question # 13 of 30 (Start time: 09:25:41 AM, 26 June 2021)

Range of the function is

$$f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$$

Select the correct option

- ○ $(0, +\infty)$
- ○ $(-\infty, 0)$
- ● $[0, +\infty)$
- ○ $(-\infty, +\infty)$



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Question # 14 of 30 (Start time: 09:25:55 AM, 26 June 2021)

Let $f(x, y)$ be a function with continuous second order partial derivatives in some circle centered at a critical point (x_0, y_0) and, let $D = f_{xx}(x_0, y_0) f_{yy}(x_0, y_0) - [f_{xy}(x_0, y_0)]^2$

Select the correct option

○	$f(x, y)$ has a relative minimum at (x_0, y_0).
○	No conclusion can be drawn.
●	$f(x, y)$ has a saddle point at (x_0, y_0).
○	$f(x, y)$ has a relative maximum at (x_0, y_0).

Click to Sa

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Quiz Start Time: 09:2

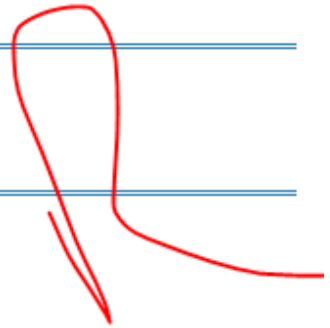
Question # 15 of 30 (Start time: 09:26:24 AM, 26 June 2021)

$x = 3\cos\theta,\ y = 3\sin\theta;\ 0 \leqslant \theta \leqslant \pi$, is a _____ form of a curve in the plane.

Select the correct option

Reloa

- parametric vector
- parametric
- explicit
- implicit



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Question # 16 of 30 (Start time: 09:26:46 AM, 26 June 2021)

If $w = f(x, y)$ where $x = g(r, s)$ then $\frac{\partial w}{\partial r} =$

Select the correct option

- $\frac{\partial w}{\partial r} = \frac{dw}{dx}\frac{\partial x}{\partial r} + \frac{dw}{dy}\frac{\partial y}{\partial r}$
- $\frac{\partial w}{\partial r} = \frac{dw}{dy}\frac{\partial y}{\partial r} + \frac{dw}{dx}\frac{\partial x}{\partial s}$
- $\frac{\partial w}{\partial r} = \frac{dw}{ds}\frac{\partial s}{\partial r} + \frac{dw}{dy}\frac{\partial y}{\partial r}$
- $\frac{\partial w}{\partial r} = \frac{dw}{dx}\frac{\partial x}{\partial w} - \frac{dw}{dy}\frac{\partial y}{\partial r}$

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Question # 17 of 30 (Start time: 09:27:03 AM, 26 June 2021)

$x - coordinate\ of\ vertex\ of\ parabola\ y = 3x^2 + 6x + 8\ is$

Select the correct option

- (●) -1
- ( ) 1
- ( ) 3
- ( ) 6

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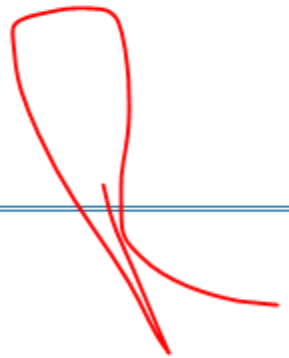
Question # 18 of 30 (Start time: 09:27:17 AM, 26 June 2021)

Length or magnitude of a unit vector is--------.

Select the correct option

- 0
- 1



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Question # 19 of 30 (Start time: 09:27:35 AM, 26 June 2021)

Which of the following is the representation of spherical coordinates

Select the correct option

- ○ (r, θ, z)
- ● (ρ, θ, φ)
- ○ (ρ, φ, z)
- ○ (x, y, z)

Click to Save An

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Quiz Start Time: 09:21 AM, 26

Question # 20 of 30 (Start time: 09:27:51 AM, 26 June 2021) Tot

Position of any point in the plane can be obtained by the two perpendicular lines known as x and y axes and together we call it as _____ coordinates system.

Select the correct option

- ○ Spherical
- ○ Rectangular
- ○ Cylindrical
- ● Cartesian

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Question # 21 of 30 (Start time: 09:28:07 AM, 26 June 2021)

Cylindrical coordinate system is used to determine the position of any object in

Select the correct option

- one dimension
- three dimension
- two dimension
- any dimension

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Question # 22 of 30 (Start time: 09:28:22 AM, 26 June 2021)

Which coordinate system uses two distances and one angle?

Select the correct option

- (●) cylindrical
- ( ) spherical
- ( ) rectangular
- ( ) none of these

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Question # 23 of 30 (Start time: 09:28:49 AM, 26 June 2021)

If $w = f(x)$ where $x = g(r, s)$ then $\frac{\partial w}{\partial s} =$

Select the correct option

- $\frac{\partial w}{\partial s} = \frac{dw}{ds}\frac{\partial s}{\partial s}$
- $\frac{\partial w}{\partial s} = \frac{dw}{dx}\frac{\partial x}{\partial s}$
- $\frac{\partial w}{\partial s} = \frac{dw}{dr}\frac{\partial r}{\partial s}$
- $\frac{\partial w}{\partial s} = \frac{dw}{ds}\frac{\partial s}{\partial r}$

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Question # 24 of 30 (Start time: 09:29:09 AM, 26 June 2021)

Partial derivative of

$$f(x,y) = x^2 + y^2 + xy$$

with respect to y is

Select the correct option

○	$x^2 + 2y + xy$
○	$x^2 + 2y + x$
●	$2y + x$
○	None of these

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Question # 25 of 30 (Start time: 09:29:26 AM, 26 June 2021)

In rectangular coordinate system, the point (1, 5) lies in

Select the correct option

- ○ fourth quadrant
- ● first quadrant
- ○ third quadrant
- ○ second quadrant

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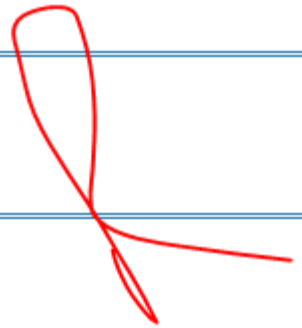
Question # 26 of 30 (Start time: 09:29:41 AM, 26 June 2021)

The differentiable functions are continuous ..... in its domain.

Select the correct option

- ○ nowhere
- ● everywhere
- ○ somewhere
- ○ only at (0,0)



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Question # 27 of 30 (Start time: 09:29:57 AM, 26 June 2021)

Double integral is used to find the ------ of solid.

Select the correct option

- ○ width
- ○ none of these
- ● volume
- ○ surface

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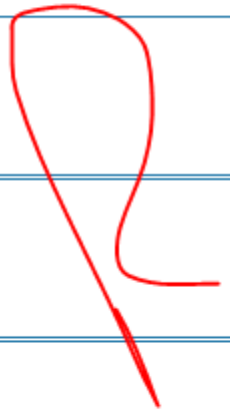
Question # 28 of 30 (Start time: 09:30:11 AM, 26 June 2021)

Value of the function f(x, y, z) = xy + z at the point (0, 1, 1) is _____.

Select the correct option

- ○ 2
- ● 1
- ○ 3
- ○ 0



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Question # 29 of 30 (Start time: 09:30:26 AM, 26 June 2021)

Value of the function $f(x,y,z) = xy + z$ *at the* point $(1, x^3, y^2)$

Select the correct option

- $y + z$
- $x^3 + y^2$
- $xy + z^2$
- $x^3 + z$

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Quiz Start Time

Question # 30 of 30 (Start time: 09:30:44 AM, 26 June 2021)

The function of the form $f(x, y) = 3x^2x^5$ is _____ in the domain.

Select the correct option

- discontinuous
- may or may not be continuous
- continuous everywhere
- piecewise continuous

Click to Save Answer & M

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